Connecting Low-Level and High-Level Visions: A Joint Optimization for Hyperspectral Image Super-Resolution and Target Detection

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***Abstract*— Typical high-level vision tasks in hyperspectral image (HSI) processing, such as target detection, often suffer from insufficient information inherent in real-world sampled data. Super-resolution, a powerful tool in HSI low-level vision, is expected to enhance the accuracy of detection results by computationally providing the high-resolution HSI (HR-HSI) with additional information. However, existing solutions for HSI super-resolution and target detection have always been implemented independently. This conventionally adopted paradigm overlooks the interconnectedness between low-level and high-level visions, inevitably introducing additional errors, redundancies, and inefficiencies. To address this challenge, in this study, we put our efforts into exploring the uncharted continent of hyperspectral remote sensing, that is, realizing the mutual guidance and joint optimization of HSI super-resolution and target detection concurrently within a unified framework. Technically, we first construct different spectral bases to span the target and background subspaces of the underlying HR-HSI. Then, we look in-depth at the intrinsic properties of the HSI tensor, henceforth jointly optimizing both tasks by innovatively developing a novel low-cubic-rank tensor approximation model with a unique constrained energy minimization (CEM) loss. While we have developed efficient algorithms to optimize the proposed model, we also put into place a refinement procedure for spectral bases, aimed at further enhancing the spectral fidelity of the fused results and the compact representation of the target subspace. Finally, empirical studies conducted on synthetic and real-world datasets substantiate that compared with state-of-the-art solutions, the proposed method delivers highly competitive and practical performance in terms of both tasks. Source codes are available at https://github.com/CX-He/HySRTD.git.**

***Index Terms*— Constrained energy minimization (CEM), hyperspectral image (HSI) super-resolution, HSI target detection, low-rank tensor representation.**

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I. Introduction

HUNDREDS of continuous and narrow bands ranging from the visible light to infrared spectrum are contained in the hyperspectral image (HSI), enabling a growing number of multifarious high-level vision tasks (e.g., target detection [1], [2], [3], [4], land-cover classification [5], [6], [7], and change detection [8], [9]) in remote sensing. However, in real-world HSI imaging, there is always an insurmountable trade-off between spatial finesse and spectral coverage [10]. In other words, obtaining a high-resolution HSI (HR-HSI) that excels in both spatial and spectral domains directly is challenging. Relying exclusively on hardware advancements to collect higher-quality data is extremely uneconomical, impractical, and unsustainable. Therefore, as a typical low-level vision tool, HSI super-resolution computationally fuses the low-resolution HSI (LR-HSI) and high-resolution multispectral image (HR-MSI), provides an appealing algorithmic alternative to obtain the HR-HSI. Mathematically, the HSI super-resolution problem can be formulated as follows:

$$\begin{cases} \mathcal{X} = \mathcal{D}_{\text{spa}}(\mathcal{Z}) + \mathcal{E}_{\mathcal{X}} \\ \mathcal{Y} = \mathcal{D}_{\text{spe}}(\mathcal{Z}) + \mathcal{E}_{\mathcal{Y}} \end{cases} \tag{1}$$

where $\mathcal{D}_{\text{spa}}(\bullet)$ is a composite operator of spatial blurring and downsampling, which degrades the third-order HR-HSI tensor $\mathcal{Z} \in \mathbb{R}^{M\times N\times L}$ to its LR-HSI representation $\mathcal{X} \in \mathbb{R}^{m\times n\times L}$ ($m \ll M$, $n \ll N$); $\mathcal{D}_{\text{spe}}(\bullet)$ denotes a spectral downsampling procedure, which can be specified as $\mathcal{Z}\times_3\mathbf{S}$, where $\times_3$ is the tensor mode-3 product and $\mathbf{S} \in \mathbb{R}^{l\times L}$ is the spectral response of the multispectral sensor mapping $\mathcal{Z}$ to its HR-MSI representation $\mathcal{Y} \in \mathbb{R}^{M\times N\times l}$ ($l \ll L$); $\mathcal{E}_{\mathcal{X}} \in \mathbb{R}^{m\times n\times L}$ and $\mathcal{E}_{\mathcal{Y}} \in \mathbb{R}^{M\times N\times l}$ represent random modeling errors associated with spatial and spectral degradation, respectively. Essentially, restoring an approximate low-rank $\mathcal{Z}$ from the collected $\mathcal{X}$ and $\mathcal{Y}$ is a challenging ill-posed inverse problem [11].

Meanwhile, high-quality HR-HSI data is a crucial prerequisite for many high-level vision tasks [12]. For example, the HSI target detection task, which plays a vital role in many

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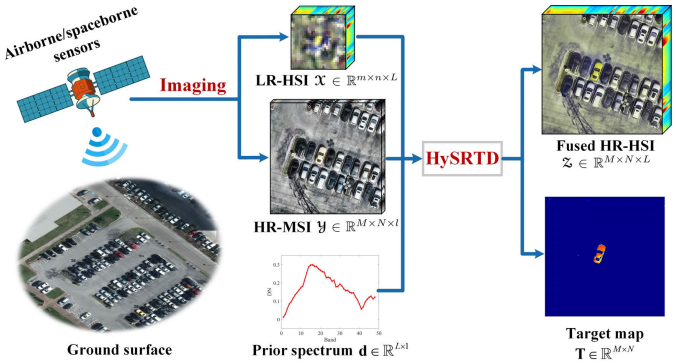


Fig. 1. Application scene of the proposed method.

remote sensing application scenarios such as national defense, criminal investigation, and environmental monitoring, depends heavily on the data quality [13]. When HR-HSI is unavailable in these scenarios, directly using the information-limited LR-HSI as a substitute will cause severe performance degradation due to the presence of mixed pixels. Mathematically, given a prior spectral signature $\mathbf{d} \in \mathbb{R}^{L\times 1}$ selected from a spectra library according to the ground material of interest and a high-quality HR-HSI $\mathcal{Z}$, the HSI target detection problem can be formulated as the following decomposition form:

$$\mathcal{Z} = \mathcal{T} + \mathcal{B} \tag{2}$$

where $\mathcal{T} \in \mathbb{R}^{M\times N\times L}$ and $\mathcal{B} \in \mathbb{R}^{M\times N\times L}$ represent the target part and background part, respectively. Problem (2) can be regarded as a binary classification problem, where $\mathcal{T}$ and $\mathcal{B}$ are distinguished from $\mathcal{Z}$ by a given $\mathbf{d}$. Then, a target map $\mathbf{T} \in \mathbb{R}^{M\times N}$ can be obtained by a thresholding operation.

Recently, many efforts have been made on the HSI super-resolution task, such as model optimization [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35] and deep learning-based [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49] methods. Meanwhile, studies on the HSI target detection have also caught a considerable amount of attention [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67]. Essentially, we anticipate that obtaining high-quality HR-HSI through low-level super-resolution methods can alleviate the interference of mixed spectral pixels and provide richer information, thereby enhancing the performance of high-level target detection methods [68]. However, it is important to note that the primary design purpose of low-level vision methods is typically to enhance the perceptual quality of visual data. Such enhancements, optimized to please the human eye, may not directly align with the feature extraction and analysis requirements of high-level vision tasks in real-world applications. Furthermore, a critical issue arises from the fact that HSI super-resolution and target detection methods are conventionally performed independently, lacking a robust connection. This fragmented practice can introduce additional errors, redundancies, and inefficiencies during the optimization process, leading to unpredictable impacts.

Motivated by the above concerns, in this study, we focus on an intriguing and highly practical problem, namely how to obtain the HR-HSI $\mathcal{Z}$ and distribution $\mathbf{T}$ of the target of interest synchronously rather than sequentially solely from the undersampled data pairs of LR-HSI $\mathcal{X}$ and HR-MSI $\mathcal{Y}$. To this end, a unified method for HSI super-resolution and target detection, named HySRTD, is proposed to jointly improve the fusion accuracy and detection rate. The application scenario is illustrated in Fig. 1. Concretely, we first dive into the spectral subspaces of HSI and look in-depth at the intrinsic properties underlying the spectral information (spectral bases) and the corresponding spatial information (representations). Then, we figure out a simple yet effective solution to separate target $\mathcal{T}$ and background $\mathcal{B}$ within the super-resolution process: spanning the target subspace by the given prior spectrum $\mathbf{d}$ and its adjacent spectral bases. Next, different from previous HSI super-resolution studies that generally borrowed off-the-shelf low-rank regularizations, we propose a novel complexity measure for third-order tensors, termed tensor cubic rank, to regularize the representation tensors. This measure alleviates the inflexible blemish of previous tensor low-rank regularizations in characterizing implicit correlations along different modes of HSI. Furthermore, we rephrase the well-known constrained energy minimization (CEM) detector [50], [51], [52], [53] as a new loss term. This CEM loss is incorporated into an objective function and jointly minimized with the above low-cubic-rank regularization. Finally, with the iterative refinement and correction of the spectral bases, both the HSI super-resolution and target detection tasks are achieved in a unified view rather than being implemented stepwise.

In short, the main contributions of this study are threefold.

1) To the best of our knowledge, this is the first attempt to jointly optimize the HSI super-resolution and target detection tasks. By carefully connecting low-level and high-level visions within a unified framework, the additional errors, redundancies, and inefficiencies introduced by the previous paradigm of independent implementations are suppressed more effectively.
2) We propose a novel tensor cubic rank that offers a more flexible multidimensional characterization of the high-order low-rank properties of the HSI's representation tensor in the transform domain. This new rank definition more flexibly considers implicit dependencies along the spatial mode, self-similarity along the non-local mode, and spectral subdivision along the spectral mode to ensure the results of the super-resolution task.
3) A novel CEM loss is designed to minimize the energy response of the background tensor, thereby facilitating a better separation of the target and background subspaces during the joint optimization procedure. This endeavor not only promotes the results of the target detection task but also robustly validates the utility of reinterpreting well-established theories from new perspectives.

The remainder of this article is organized as follows. Section II reviews the literature and presents the preliminaries. Section III elaborates the main theoretical results of the proposed method. Experimental evaluations and ablation studies

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are carried out in Sections IV and V, respectively. Finally, the conclusions and future work are summarized in Section VI.

II. Related Work and Preliminaries

A. Related Work

1) HSI Super-Resolution: Recent progress in model optimization-based HSI super-resolution methods is mainly built on sparse representation, matrix factorization, and tensor representation theories. The first family assumes that HSI can be over-completely represented by a certain dictionary. Hence, image fusion is achieved by introducing certain priors to learn a corresponding sparse coding, e.g., non-negativity [14], Bayesian inference [15], [16], [69], low-rank prior [17], [18], and self-similarity [19], [20]. However, their performance heavily depends on the specific design of the dictionary, especially when handling datasets with wide spectral coverage and heterogeneous spatial texture. The second family considers spectral and spatial information separately when establishing the optimization model. The estimation of HR-HSI is achieved by learning a spectral matrix and a corresponding coefficient matrix with various priors [21], [22], [23], [24], [25]. However, these methods require flattening the HSI tensor into a matrix, which inevitably destroys the structural correlations of the HSI tensor.

To tackle this issue, various tensor representation methods have received extensive attention. Studies [26], [27] suggested describing the high-order correlations of HSI through canonical polyadic (CP) decomposition, but the solution of CP rank is hard to compute. Based on Tucker decomposition, studies [28], [29] proposed the coupled sparse and weighted low-rank tensor representation models. However, the approximation of Tucker rank essentially requires unfolding the HSI tensor along each mode. Ring decomposition [31], [32], [33] uses a set of core tensors with circular dimensions to represent the latent tensor, further improving the flexibility of Train decomposition [30] and delivering more promising HSI super-resolution results. However, the overlarge computational requirements limit the real application of Ring decomposition. As a higher-order generalization of matrix singular value decomposition (SVD), tensor-SVD (t-SVD) decomposes a third-order tensor into two orthogonal tensors and a f-diagonal tensor based on the closed tensor-tensor product (t-product) under the invertible linear transformation [70]. The high-order singular values provided by t-SVD can effectively map the intrinsic complexity of the HSI tensor [34]. Nevertheless, limited by strict dimensionality requirements, t-SVD fails to handle the different high-order correlations along different modes of HSI tensor [71], [72], while the yielded tubal rank [73], [74] also neglects to measure the implicit low-rank property in the transform domain. Such inflexible and insufficient low-rank characterizations naturally lead to suboptimal HSI super-resolution results [35].

In addition, data-driven deep fusion networks have witnessed increasing interest recently [36], [37], [38], [39], [40], [41], [42]. However, acquiring large-scale high-quality labeled samples is challenging, especially in remote sensing HSI imaging [75], [76]. Some studies focus on unsupervised approaches [43], [44], [45], [46], [47], [48], [49], but the training procedures are often highly time-consuming, and the trained networks are less generalized. In Section IV, popular deep fusion networks will be included in evaluations.

2) HSI Target Detection: With high precision and low computational burden, the CEM filter [50] and its variants [51], [52], [53] have become the most popular HSI target detectors. CEM filter maximizes the detection rate of target pixels by calculating an autocorrelation matrix to minimize the energy response of background pixels. However, all pixels in HSI will participate in the calculation, which causes certain errors. Also, there are other classic detectors, e.g., orthogonal subspace projection [54], spectral matched filter [55], and adaptive consistency/cosine estimator [56]. Similar to CEM, once the spatial quality of HSI is degraded, these detectors are hard to give accurate results for datasets with complex backgrounds.

Recently, the sparse and low-rank representation models have been actively studied [57], [58], [59], [60]. Assuming that the target and background exist in different subspaces, one can build appropriate regularization models to estimate them. However, these methods' computational efficiency is generally inferior to classic spectral matching-based methods due to the complicated optimization arithmetic involved. Meanwhile, the mixed pixels will also diminish the detection rates in real applications. Deep learning methods have powerful target feature extraction capabilities [77]. On the one hand, supervised detection networks [61], [62], [63], [64] distinguish the target from background by learning the deep nonlinear features. However, they are prone to overfitting due to the scarcity of labeled spectra samples. On the other hand, unsupervised networks [65], [66], [67] employ deep autoencoder-based models to discriminate the targets and backgrounds. However, their performance also depends on the quality of input datasets, which is not always met in real-world HSI imaging. Besides, the overcomplex network structure and low implementation efficiency make it difficult for unsupervised networks to deploy effectively in practical remote sensing applications.

In summary, detection performance is highly dependent on data quality, arguing an important but underexplored connection between HSI low-level and high-level visions. Existing practice has not yet considered the connection between these two different levels of HSI processing tasks. In this study, we commit to building a bridge between the HSI super-resolution and target detection and finding a new unified solution.

B. Tensor Algebra Preliminaries

Hereinafter, the scalars, vectors, matrices, and tensors are denoted by x, $\mathbf{x}$, $\mathbf{X}$, and $\mathcal{X}$. Because we focus on using tensor representation as a tool for processing HSI, the tensors are only defined to the third order for simplicity and readability. The (i, j, k)th entry of a third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is denoted as $\mathcal{X}_{(i,j,k)}$. The column, row, and tube fibers of $\mathcal{X}$ are denoted by $\mathcal{X}_{(:,j,k)}$, $\mathcal{X}_{(i,:,k)}$, and $\mathcal{X}_{(i,j,:)}$, respectively. Meanwhile, the frontal, lateral, and horizontal slices of $\mathcal{X}$ are denoted by $\mathcal{X}_{(:,:,k)}$, $\mathcal{X}_{(:,j,:)}$, and $\mathcal{X}_{(i,:,:)}$, respectively. Specifically, the compact form $\mathbf{X}^{(i)}$ is invited to represent the ith frontal slice. The flattening matrix of $\mathcal{X}$ along ith mode is denoted as $\mathbf{X}_{(3)} \in \mathbb{R}^{n_3 \times n_1 n_2}$. The identity matrix is denoted as

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$\mathbf{I}$, and the zeros matrix is denoted as $\mathbf{O}$. The tensor Frobenius norm is defined as $\|\mathcal{X}\|_F = (\sum_{i,j,k} |\mathcal{X}_{(i,j,k)}|^2)^{(1/2)}$.

We now clarify some important tensor ingredients related to this work [70], [74], [78]. Given a third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, we consider the invertible linear transform $\mathcal{L} : \mathbb{R}^{n_1 \times n_2 \times n_3} \to \mathbb{R}^{n_1 \times n_2 \times n_3}$, which is particularly denoted as

$$\bar{\mathcal{X}} = \mathcal{L}(\mathcal{X}) = \mathcal{X} \times_3 \mathbf{L} \tag{3}$$

where $\mathbf{L} \in \mathbb{R}^{n_3 \times n_3}$ can be any invertible real matrix. Consequently, we have $\mathcal{X} = \mathcal{L}^{-1}(\bar{\mathcal{X}}) = \bar{\mathcal{X}} \times_3 \mathbf{L}^{-1}$.

Definition 1 (t-Product Under $\mathcal{L}$): [70] The t-product of $\mathcal{U} \in \mathbb{R}^{n_1 \times n \times n_3}$ and $\mathcal{S} \in \mathbb{R}^{n \times n_2 \times n_3}$ under $\mathcal{L}$ is defined as

$$\mathcal{X} = \mathcal{U} *_{\mathcal{L}} \mathcal{S} \tag{4}$$

which is equivalent to $\bar{\mathcal{X}} = \bar{\mathcal{U}} \odot \bar{\mathcal{S}}$, where operator $\odot$ denotes the frontal slice-wise product of two tensors.

Due to space limitations in the article, we have provided some preliminaries in Appendix A of the Supplementary.

Theorem 1 (t-SVD Under $\mathcal{L}$): [70] Using Definitions 1, A.1 (tensor transpose), A.2 (identity tensor), A.3 (orthogonal tensor), and A.4 (f-diagonal tensor), the t-SVD of $\mathcal{X}$ under $\mathcal{L}$ is computed as the following factorized form:

$$\mathcal{X} = \mathcal{U} *_{\mathcal{L}} \mathcal{S} *_{\mathcal{L}} \mathcal{V}^T \tag{5}$$

where $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensors, and $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an f-diagonal tensor. Indeed, it is equivalent to $\bar{\mathbf{X}} = \bar{\mathbf{U}}\bar{\mathbf{S}}\bar{\mathbf{V}}^T$ in the transform domain according to Lemma A.1.

Based on Theorem 1, studies [73], [74] propose to measure the tensor complexity by the number of non-zero singular tube fibers in $\mathcal{S}$, namely tubal rank (see Definition A.5). However, the tubal rank essentially ignores the property of singular values in the transform domain. To overcome this drawback, Wang et al. [78] suggest the following tensor average rank, where the transform matrix $\mathbf{L}$ satisfies $\mathbf{L}^T\mathbf{L} = \mathbf{L}\mathbf{L}^T = \mathbf{I}$:

Definition 2 (Tensor Average Rank): [78] Suppose $\mathcal{L}$ in (3) is a real orthogonal transform, the average rank of a given tensor $\mathcal{X}$ under $\mathcal{L}$ is defined as the average across the ranks of all frontal slices in the transform domain

$$\text{Rank}_{\text{a}}(\mathcal{X}) = \frac{1}{n_3}\sum_{i=1}^{n_3} \text{Rank}\left(\bar{\mathbf{X}}^{(\text{i})}\right) = \frac{1}{n_3}\sum_{i,k} \delta\left(\bar{\mathcal{S}}_{(i,i,k)}\right) \tag{6}$$

where $\mathcal{S}$ is calculated by Theorem 1, $\delta(x)$ is the Kronecker delta function that sets x to 1 if x is greater than 0 and remains x unchanged if x is equal to 0.

The above tensor average rank achieves a more profound tensor complexity measure than tensor tubal rank in the real orthogonal transform domain. However, the limitation of orientational sensitivity of tubal rank in handling the different correlations along different modes of the HSI tensor is still unaddressed. Later in Section III-C, we define a new tensor cubic rank to more flexibly characterize the 3-D correlated HSI tensor in the transform domain.

C. CEM Filter

Consider a 2-D spectral matrix $\mathbf{X} \in \mathbb{R}^{L \times MN}$ flattened from a 3-D HSI tensor $\mathcal{X} \in \mathbb{R}^{M \times N \times L}$, each column of $\mathbf{X}$ is a spectral vector. CEM-based detectors [50], [51], [52], [53] build an optimal finite impulse response filter specified by $\mathbf{w} = [w_1, w_2, \cdots, w_L]^T$. Under the constrain that the response of the given prior target signature $\mathbf{d} \in \mathbb{R}^{L \times 1}$ to filter $\mathbf{w}$ is a constant, the CEM filter minimizes the total output energy

$$\min_{\mathbf{w}} \ \mathbf{w}^T\mathbf{R}\mathbf{w}, \quad \text{s.t. } \mathbf{w}^T\mathbf{d} = 1 \tag{7}$$

where $\mathbf{R} = (1/MN)\mathbf{X}\mathbf{X}^T$ is the maximum likelihood estimation of the autocorrelation matrix. The closed-form solution to this optimization problem can be given by

$$\hat{\mathbf{w}} = \frac{\mathbf{R}^{-1}\mathbf{d}}{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}}. \tag{8}$$

The obtained filter $\mathbf{w}$ compels the target signals to output higher energy and the background signals to output lower energy. Thus, the output response of CEM is given by

$$\mathbf{y} = \left(\hat{\mathbf{w}}\right)^T\mathbf{X} = \frac{\mathbf{d}^T\mathbf{R}^{-1}}{\mathbf{d}\mathbf{R}^{-1}\mathbf{d}^T}\mathbf{X}. \tag{9}$$

Finally, each entry of the output $\mathbf{y}$ is compared with a fixed threshold, one can be called target present if the result value is higher than the threshold. Otherwise, there is a target absent.

In real applications, however, detection rates are not always guaranteed if CEM-based methods are independently performed on the fused HR-HSI. While it is challenging to maintain spectral invariability during the fusion process, the inclusion of target samples in calculating $\mathbf{y}$ can introduce uncertainty into the detection results. These errors may be magnified when independently implementing super-resolution and target detection. In Section III-B, we rephrase the CEM filter into a novel loss term to mitigate the above issues.

III. Unified HSI Super-Resolution and Target Detection Method

In this section, we begin by presenting the fundamental design of the HySRTD framework. We then elaborate on the motivation and definition of the novel CEM loss and tensor cubic rank minimization. Subsequently, the algorithmic solution proceeds through alternating optimization. An overview of our proposed HySRTD method is depicted in Fig. 2. To enhance readability, the iteration index is only specified in the algorithm procedures and schematics.

A. Overall Framework of the Proposed HySRTD Method

We aim to reconstruct the HR-HSI $\mathcal{Z}$ while simultaneously separating its target part $\mathcal{T}$ and background part $\mathcal{B}$ from the collected data pair of LR-HSI $\mathcal{X}$ and HR-MSI $\mathcal{Y}$ within a unified framework. Therefore, by combining the degradation models in (1) and (2), one can construct the regularization model consisting of following two fidelity terms:

$$\min_{\mathcal{T},\mathcal{B}} \frac{1}{2}\left\|\mathcal{X} - \mathcal{D}_{\text{spa}}(\mathcal{T}+\mathcal{B})\right\|_F^2 + \frac{1}{2}\left\|\mathcal{Y} - \mathcal{D}_{\text{spe}}(\mathcal{T}+\mathcal{B})\right\|_F^2. \tag{10}$$

The challenge lies in regularizing $\mathcal{T}$ and $\mathcal{B}$ with appropriate priors. Previous studies on HSI target detection commonly suggest that $\mathcal{B}$ is low-rank, and $\mathcal{T}$ is sparse [58], [59]. However, adhering to this common thread does not guarantee

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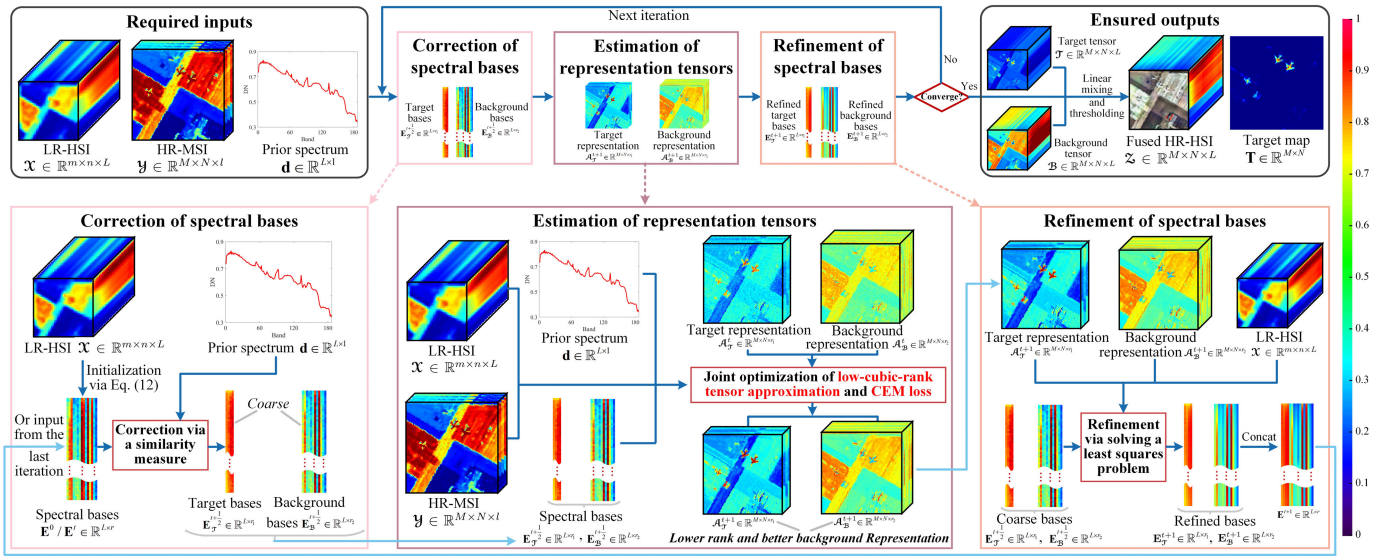


Fig. 2. Schematic of the overall framework of the proposed method.

the fusion quality for the reconstructed HR-HSI $\mathcal{Z}$ when addressing the problem in (10) with two fidelity terms.

Fortunately, the target and background components of an HSI lie in different spectral subspaces. This suggests that the spectral bases of the latent $\mathcal{Z}$ should comprise two parts: one spanning the target subspace and the other spanning the background subspace. This can be described as follows:

$$\mathcal{Z} = \mathcal{A}_{\mathcal{T}} \times_3 \mathbf{E}_{\mathcal{T}} + \mathcal{A}_{\mathcal{B}} \times_3 \mathbf{E}_{\mathcal{B}} \tag{11}$$

where $\mathcal{A}_{\mathcal{T}} \in \mathbb{R}^{M\times N\times r_1}$, $\mathbf{E}_{\mathcal{T}} \in \mathbb{R}^{L\times r_1}$, $\mathcal{A}_{\mathcal{B}} \in \mathbb{R}^{M\times N\times r_2}$, and $\mathbf{E}_{\mathcal{B}} \in \mathbb{R}^{L\times r_2}$ are the representation tensors and spectral bases of tensors $\mathcal{T}$ and $\mathcal{B}$, respectively. Consequently, the simultaneous accomplishment of both tasks converts to estimating these above variables. However, directly optimizing four variables in one model leads to high complexity and instability. Meanwhile, it is also very challenging to devise an optimization algorithm that is both effective and efficient.

Therefore, as illustrated in the first row of Fig. 2, we propose to iteratively repeat the following three procedures to build a simple yet effective HySRTD framework:

1) Correction of Spectral Bases: Ideally, $\mathbf{E}_{\mathcal{T}}$ should be the given prior spectrum. However, the information carried in one single spectral signature is limited. Therefore, we first augment $\mathbf{E}_{\mathcal{T}}$ to enhance its capability to represent the target subspace. As the spectral information of $\mathcal{Z}$ is primarily contained in $\mathcal{X}$, one can initialize a coarse spectral information $\mathbf{E} \in \mathbb{R}^{L\times r}$ by

$$\mathbf{E} = \ell_{\text{VCA}}(\mathcal{X}) \tag{12}$$

where $\ell_{\text{VCA}}(\bullet)$ denotes the well-known vertex component analysis function [79], and $r = r_1 + r_2$. What is noteworthy is that it is unnecessary to search for an optimal $\mathbf{E}$ during the initialization phase, because $\mathbf{E}$ will be refined later to further ensure the performance of the super-resolution task.

Subsequently, we initialize $\mathbf{E}_{\mathcal{T}} = \mathbf{d}$ and $\mathbf{E}_{\mathcal{B}} = \emptyset$, correcting them with the assistance of the spectral information $\mathbf{E}$ by

$$\begin{cases} \mathbf{E}_{\mathcal{T}} = [\mathbf{E}_{\mathcal{T}}, \mathbf{E}(:,i)], & \text{if } \ell_{\text{SAM}}(\mathbf{d}, \mathbf{E}(:,i)) \le \eta \\ \mathbf{E}_{\mathcal{B}} = [\mathbf{E}_{\mathcal{B}}, \mathbf{E}(:,i)], & \text{otherwise} \end{cases} \tag{13}$$

where $i \in [2, r]$, $\ell_{\text{SAM}}(\mathbf{x}_1, \mathbf{x}_2)$ denotes the spectral angle mapper (SAM) function [80] used to measure the adjacent degree of two spectral vectors, and η is a positive radian.

Remark 1: The advantage of the above correction process is evident: at the beginning of each iteration, the target subspace is consistently spanned by the prior spectrum $\mathbf{d}$ and its adjacent signatures. In other words, this correction ensures the effectiveness of HSI target detection by keeping the spectral bases of target and background subspaces separated.

2) Estimation of Representation Tensors: With the corrected spectral bases of high-resolution $\mathcal{T}$ and $\mathcal{B}$, we then build the following regularization model converted from (10) to estimate the representation tensors $\mathcal{A}_{\mathcal{T}}$ and $\mathcal{A}_{\mathcal{B}}$:

$$\begin{aligned} \min_{\mathcal{A}_{\mathcal{T}}, \mathcal{A}_{\mathcal{B}}} & \frac{1}{2}\left\|\mathcal{X} - \mathcal{D}_{\text{spa}}(\mathcal{A}_{\mathcal{T}}\times_3\mathbf{E}_{\mathcal{T}} + \mathcal{A}_{\mathcal{B}}\times_3\mathbf{E}_{\mathcal{B}})\right\|_F^2 \\ & + \frac{1}{2}\left\|\mathcal{Y} - \mathcal{D}_{\text{spe}}(\mathcal{A}_{\mathcal{T}}\times_3\mathbf{E}_{\mathcal{T}} + \mathcal{A}_{\mathcal{B}}\times_3\mathbf{E}_{\mathcal{B}})\right\|_F^2 \\ & + \lambda\Phi_{\text{SR}}\left([\mathcal{A}_{\mathcal{T}}, \mathcal{A}_{\mathcal{B}}]_3\right) + \gamma\Phi_{\text{TD}}(\mathcal{A}_{\mathcal{B}}\times_3\mathbf{E}_{\mathcal{B}}) \end{aligned} \tag{14}$$

where λ and γ denote the regularization parameters, and $[\mathcal{A}_{\mathcal{T}}, \mathcal{A}_{\mathcal{B}}]_3 \in \mathbb{R}^{M\times N\times r}$ denotes the concatenated tensor along the third mode, which can be regarded as the global representation tensor of the latent HR-HSI $\mathcal{Z}$.

The first regularization, $\Phi_{\text{SR}}(\bullet)$ in (14), is applied to the global representation tensor $[\mathcal{A}_{\mathcal{T}}, \mathcal{A}_{\mathcal{B}}]_3$ to ensure the performance of super-resolution. A feasible choice for $\Phi_{\text{SR}}(\bullet)$ is the tensor average rank $\text{Rank}_{\text{a}}(\bullet)$, as defined in Definition 2. However, as analyzed earlier, $\text{Rank}_{\text{a}}(\bullet)$ is not tailor-made for the HSI tensor, i.e., it does not flexibly consider the spatial redundancy, non-local self-similarity, and spectral subdivision that HSI possesses along different modes in the spectral subspace. The second regularization, $\Phi_{\text{TD}}(\bullet)$, applied to the background tensor $\mathcal{A}_{\mathcal{B}}\times_3\mathbf{E}_{\mathcal{B}}$, is responsible for background isolation. In model-based target detection studies, low-rank regularization is often employed as $\Phi_{\text{TD}}(\bullet)$; however, this practice cannot simultaneously ensure the performance of super-resolution. In statistically based studies,

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the correlations between low-level and high-level visions are often neglected, introducing additional errors since they are independently implemented after the super-resolution process. In Sections III-B and III-C, we will reconsider these dilemmas from a high-dimensional tensor representation perspective.

Remark 2: Compared with the previous studies, a notable profit of estimating representation tensors via the model in (14) is the further simplification of modeling. In other words, only two variables are required to be optimized to obtain the target tensor $\mathcal{T}$, background tensor $\mathcal{B}$, and HR-HSI tensor $\mathcal{Z}$. Meanwhile, two regularization terms responsible for two tasks have been judiciously imposed after correcting the spectral bases $\mathbf{E}_{\mathcal{T}}$ and $\mathbf{E}_{\mathcal{B}}$ in each iteration. This breaks the limitation of previous studies that were unable to optimize two tasks simultaneously and further lays the groundwork for the success of both tasks.

The optimization for solving (14) is discussed in Section III-D after $\Phi_{\text{SR}}(\bullet)$ and $\Phi_{\text{TD}}(\bullet)$ are explicitly determined.

3) Refinement of Spectral Bases: Once the representations $\mathcal{A}_{\mathcal{T}}$ and $\mathcal{A}_{\mathcal{B}}$ are estimated, we consider further refining the spectral bases by solving the following least squares problem:

$$\min_{\mathbf{E}_{\mathcal{T}},\mathbf{E}_{\mathcal{B}}} \frac{1}{2}\left\|\mathcal{X}-\mathcal{D}_{\text{spa}}(\mathcal{A}_{\mathcal{T}}\times_3\mathbf{E}_{\mathcal{T}}+\mathcal{A}_{\mathcal{B}}\times_3\mathbf{E}_{\mathcal{B}})\right\|_F^2 + \mu\|[\mathbf{E}_{\mathcal{T}},\mathbf{E}_{\mathcal{B}}]\|_F^2 \quad (15)$$

whose closed-form solution is given by

$$\begin{cases}\hat{E}_{\mathcal{T}}=\left(\mathbf{X}_{(3)}\mathbf{D}_{\mathcal{T}}^T\right)\left(\mathbf{D}_{\mathcal{T}}\mathbf{D}_{\mathcal{T}}^T+\mu\mathbf{I}\right)^{-1}\\ \hat{E}_{\mathcal{B}}=\left(\mathbf{X}_{(3)}\mathbf{D}_{\mathcal{B}}^T\right)\left(\mathbf{D}_{\mathcal{B}}\mathbf{D}_{\mathcal{B}}^T+\mu\mathbf{I}\right)^{-1}\end{cases} \quad (16)$$

where $\mathbf{D}_{\mathcal{T}}=(\mathcal{D}_{\text{spa}}(\mathcal{A}_{\mathcal{T}}))_{(3)}$ and $\mathbf{D}_{\mathcal{B}}=(\mathcal{D}_{\text{spa}}(\mathcal{A}_{\mathcal{B}}))_{(3)}$. After $\mathbf{E}_{\mathcal{T}}$ and $\mathbf{E}_{\mathcal{B}}$ are refined, we can prepare a refined $\mathbf{E}$ as input to the correction procedure in (13) of next iteration

$$\mathbf{E}=[\mathbf{E}_{\mathcal{T}},\mathbf{E}_{\mathcal{B}}]. \quad (17)$$

Remark 3: The benefit of the above design is that each refinement learns better spectral bases than their initial values, further enhancing super-resolution accuracy and providing robust spectral representation for target separation. Indeed, (15) involves an overdetermined system ($r \ll MN$), enabling straightforward regularization through the Frobenius norm to obtain robust spectral bases. Meanwhile, the fidelity term related to spectral degradation is not included in the refinement, reducing the solution's difficulty and complexity.

The complete procedure of the overall framework of the proposed HySRTD method is summarized in Algorithm 1, where the functional energy used to control the iteration is $\|\mathcal{Z}^{t+1}-\mathcal{Z}^t\|_F^2/\|\mathcal{Z}^t\|_F^2$. Finally, if the energy response of a pixel in $\mathcal{T}^{t+1}$ is larger than a fixed threshold, it is claimed to be a target pixel [51]. In Sections III-B and III-C, our focus is on designing robust regularizers for the model in (14).

Algorithm 1 HySRTD for HSI Super-Resolution and Target Detection

Require: LR-HSI $\mathcal{X}$, HR-HSI $\mathcal{Y}$, prior spectrum $\mathbf{d}$.

1: **Initialization:** Estimate $\mathbf{E}^0$ via Eq. (12); Set number of spectral bases r, regularization parameter $\mu = 10^{-4}$, threshold $\eta = 0.1$, iteration $t = 0$, maximum iteration T, stop criterion ε.
2: **while** not converged **do**
3: **Initialization:** Set $\mathbf{E}_{\mathcal{T}}=\mathbf{d}$, $\mathbf{E}_{\mathcal{B}}=\emptyset$.
4: **Correction of spectral bases:** Update $\mathbf{E}_{\mathcal{T}}^{t+\frac{1}{2}}$, $\mathbf{E}_{\mathcal{B}}^{t+\frac{1}{2}}$ using $\mathbf{E}^t$ via Eq. (13).
5: **Estimation of representation tensors:** Update $\mathcal{A}_{\mathcal{T}}^{t+1}$ and $\mathcal{A}_{\mathcal{B}}^{t+1}$ by solving Eq. (14), and compute $\mathcal{T}^{t+1}$ by $\mathcal{A}_{\mathcal{T}}^{t+1}\times_3\mathbf{E}_{\mathcal{T}}^{t+\frac{1}{2}}$.
6: **Refinement of spectral bases:** Update $\mathbf{E}_{\mathcal{T}}^{t+1}$ and $\mathbf{E}_{\mathcal{B}}^{t+1}$ via Eq. (16), and compute $\mathbf{E}^{t+1}$ via Eq. (17).
7: **Reconstruction of HR-HSI:** Compute $\mathcal{Z}^{t+1}$ via Eq. (11).
8: Check the convergence: $t > T$ or functional energy $< \varepsilon$.
9: Update iterations: $t = t + 1$.
10: **end while**

Ensure: HR-HSI $\mathcal{Z}^{t+1}$; Target tensor $\mathcal{T}^{t+1}$.

B. Novel CEM Loss Regularization

As the previous analysis indicates, CEM is not well-suited for the considered application scenarios. Our focus in this study turns to the concurrent optimization of both HSI target detection and super-resolution tasks through the model presented in (14). Consequently, instead of sequentially borrowing the CEM detector after completing the super-resolution process, we rephrase it as the following novel regularization:

Proposition 1: For any third-order HSI tensor $\mathcal{X} \in \mathbb{R}^{M\times N\times L}$, we have the following regularization term defined as the sum of all CEM responses of all tube fibers in $\mathcal{X}$ with respect to a given spectral signature $\mathbf{d}$, namely, the CEM loss:

$$\Phi_{\text{CEMloss}}(\mathcal{X})=\sum_{i,j}\left(\frac{(\mathbf{R}+\beta\mathbf{I})^{-1}\mathbf{d}}{\mathbf{d}^T(\mathbf{R}+\beta\mathbf{I})^{-1}\mathbf{d}}\right)^T\mathcal{X}_{(i,j,:)} =\text{Tr}\left(\frac{\text{Re}(\mathbf{d})^T(\mathbf{R}+\beta\mathbf{I})^{-1}}{\mathbf{d}(\mathbf{R}+\beta\mathbf{I})^{-1}\mathbf{d}^T}\mathbf{X}_{(3)}\right) \quad (18)$$

where $\text{Re}(\mathbf{d})^T \in \mathbb{R}^{L\times MN}$ is the replication of vector $\mathbf{d}$ by MN times, $\text{Tr}(\bullet)$ is the trace of matrix, and β is a very small positive number.

The autocorrelation matrix $\mathbf{R}$ here may become ill-conditioned and non-invertible due to the high dimensionality of HSI tensor $\mathcal{X}$. Therefore, a scaled identity matrix $\beta\mathbf{I}$ is added to ensure the stability of CEM loss (i.e., Tikhonov regularization).

Remark 4: In the HSI target detection task, pixels of the background tensor $\mathcal{B}$ should exhibit lower CEM responses than $\mathcal{Z}$; hence, each tube fiber in $\mathcal{B}$ should have a CEM response close to 0. In other words, the CEM loss value of $\mathcal{B}$ should be very small. Accordingly, we impose the proposed CEM loss $\Phi_{\text{CEMloss}}(\bullet)$ to $\mathcal{B}=\mathcal{A}_{\mathcal{B}}\times_3\mathbf{E}_{\mathcal{B}}$ as $\Phi_{\text{TD}}(\bullet)$ in (14) to better distinguish the target and background tensors during the iterative optimization process. Note that imposing the CEM loss on the latent background tensor instead of the reconstructed HR-HSI further prevents the occurrence of pathological detection errors caused by the participation of target pixels in calculating the overall energy response.

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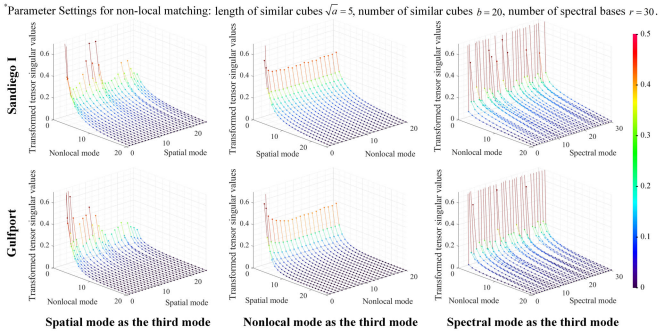


Fig. 3. Distributions of mean tensor singular values in the transform domain of all constructed tensors when different modes are rearranged as the third mode.

C. Novel Tensor Low-Cubic-Rank Regularization

According to the definition in (6), tensor average rank considers the inter-frontal slice correlations of a given third-order tensor along the third mode through transform $\mathcal{L}$, while it takes into account the intra-frontal slice correlation along the first two modes through SVD in the transform domain. However, the resulting orientational sensitivity makes it challenging to flexibly handle HSI data that exhibits different correlations along different modes. To illustrate, we examine two datasets, Sandiego I and Gulfport (see Section IV-A for details), as examples to observe these correlations closely. After estimating their spectral bases via (12), we solve a simple Frobenius norm regularized minimization problem to estimate their representation tensors. The solutions can be approximately considered as the spectral subspace representations of HR-HSI. This is supported by the reconstruction results, which easily achieve a peak signal-to-noise ratio (PSNR) result greater than 45 dB. Subsequently, we perform nonlocal matching on the representation tensor and conduct t-SVD under $\mathcal{L}$ by rearranging all different modes of the constructed tensors into the third mode. As depicted in Fig. 3, regardless of which mode of the constructed tensors is rearranged as the third mode, the mean tensor singular values in the transform domain of all constructed tensors exhibit a rapid decrease toward zero. However, their respective magnitudes and patterns are evidently different.

The above observations indicate that all three rearrangement modes should be considered individually when exploiting the low average rank properties of constructed tensors. Bearing this indication in mind, we have a new tensor complexity measure defined as follows:

Definition 3 (Tensor Cubic Rank): Suppose $\mathcal{L}$ in (3) is a real orthogonal transform, the tensor cubic rank of a given tensor $\mathcal{X}\in\mathbb{R}^{n_1\times n_2\times n_3}$ under $\mathcal{L}$ is defined as the following sum across the average ranks of the mode-j rearranged tensors $\vec{\mathcal{X}^j}$, $j=[1,3]$ in the transform domain:

$$\mathrm{Rank_c}(\mathcal{X})=\sum_{j=1}^{3}\omega_j\mathrm{Rank_a}\left(\vec{\mathcal{X}^j}\right)=\sum_{j=1}^{3}\frac{\omega_j}{n_j}\sum_{i,k}\delta\left(\bar{\mathbf{s}}^{\vec{j}}_{(i,i,k)}\right) \quad (19)$$

where $\mathcal{S}^{\vec{j}}$ is calculated by $\vec{\mathcal{X}^j}=\mathcal{U}^{\vec{j}}*_{\mathcal{L}}\mathcal{S}^{\vec{j}}*_{\mathcal{L}}(\mathcal{V}^{\vec{j}})^T$, and ω_j are non-negative weights assigned to $\vec{\mathcal{X}^j}$.

Remark 5: We emphasize the significance of the above new tensor rank definition for measuring the intrinsic correlations of tensorized HSI data. On the one hand, it alleviates the orientational sensitivity drawback of tensor average rank by flexibly measuring the sparsity of transformed singular values for HSI tensors with different rearrangement patterns. This also further allows for a more comprehensive consideration of correlations in the representation tensor of HR-HSI along different modes. On the other hand, it additionally takes into account the respective importance of correlations along each mode through a reweighting strategy, thereby facilitating more accurate low average rank tensor characterization results.

Let $\mathcal{R}_i(\bullet)$ be a linear transform that maps a third-order tensor $\mathcal{X}$ into i self-correlated constructed tensors $\mathcal{X}_i$. For any $\beta>0$, we have the following low-cubic-rank tensor approximation:

$$\Phi_{\mathrm{SR}}(\mathcal{X})=\min_{\mathcal{X}_i}\sum_i\left(\frac{\beta}{2}\|\mathcal{R}_i(\mathcal{X})-\mathcal{X}_i\|_F^2+\mathrm{Rank_c}(\mathcal{X}_i)\right). \quad (20)$$

In summary, the explicit forms of $\Phi_{\mathrm{CEMloss}}(\mathcal{X})$ in Propositions 1 and $\Phi_{\mathrm{SR}}(\mathcal{X})$ in Definition 3 ensure that the estimation of representations $\mathcal{A}_{\mathcal{T}}$ and $\mathcal{A}_{\mathcal{B}}$ in (14) can be converted as the following separable multivariables problem:

$$\begin{aligned}\min_{\mathcal{A}_{\mathcal{T}},\mathcal{A}_{\mathcal{B}},\mathcal{A}_i}\ &\frac{1}{2}\left\|\mathcal{X}-\mathcal{D}_{\mathrm{spa}}(\mathcal{A}_{\mathcal{T}}\times_3\mathbf{E}_{\mathcal{T}}+\mathcal{A}_{\mathcal{B}}\times_3\mathbf{E}_{\mathcal{B}})\right\|_F^2\\&+\frac{1}{2}\left\|\mathcal{Y}-\mathcal{D}_{\mathrm{spe}}(\mathcal{A}_{\mathcal{T}}\times_3\mathbf{E}_{\mathcal{T}}+\mathcal{A}_{\mathcal{B}}\times_3\mathbf{E}_{\mathcal{B}})\right\|_F^2\\&+\lambda\sum_i\left(\frac{\beta}{2}\left\|\mathcal{R}_i\left([\mathcal{A}_{\mathcal{T}},\mathcal{A}_{\mathcal{B}}]_3\right)-\mathcal{A}_i\right\|_F^2+\mathrm{Rank_c}(\mathcal{A}_i)\right)\\&+\gamma\,\Phi_{\mathrm{CEMloss}}(\mathcal{A}_{\mathcal{B}}\times_3\mathbf{E}_{\mathcal{B}})\end{aligned} \quad (21)$$

where $\mathcal{R}_i([\mathcal{A}_{\mathcal{T}},\mathcal{A}_{\mathcal{B}}]_3)\in\mathbb{R}^{a\times b\times r}$ stands for the constructed tensor of the concatenated representation tensor for each exemplar cube at location i, $\mathcal{A}_i$ denotes the corresponding latent low-cubic-rank structure, $(a)^{1/2}$ and b are the length and number of similar cubes, and the specific process of estimating the representation tensors is illustrated in Fig. 4.

Through the integration of the suggested CEM loss and low-cubic-rank regularization into the model presented in (14), the CEM response of the underlying high-resolution background tensor is minimized while the faithful super-resolution result is ensured. Compared with the existing HSI processing study [71] that also accounts for the orientational sensitivity of the t-SVD algebraic framework, our model in (21) demonstrates clear advantages. First, we achieve a more accurate low-rank tensor approximation by evaluating the implicit sparsity of the f-diagonal tensor in the transform domain. Second, we extend our consideration to the low-rank representation in the spectral subspace, which is particularly suitable for HSI tensors exhibiting strong spectral correlations. Third, our design establishes a stronger connection between low-level and high-level visions, greatly enhancing the practical application value in real-world HSI processing.

D. Algorithmic Solution

With the regularization terms determined, solving (14) in Algorithm 1 is now equivalent to solving (21). For a program

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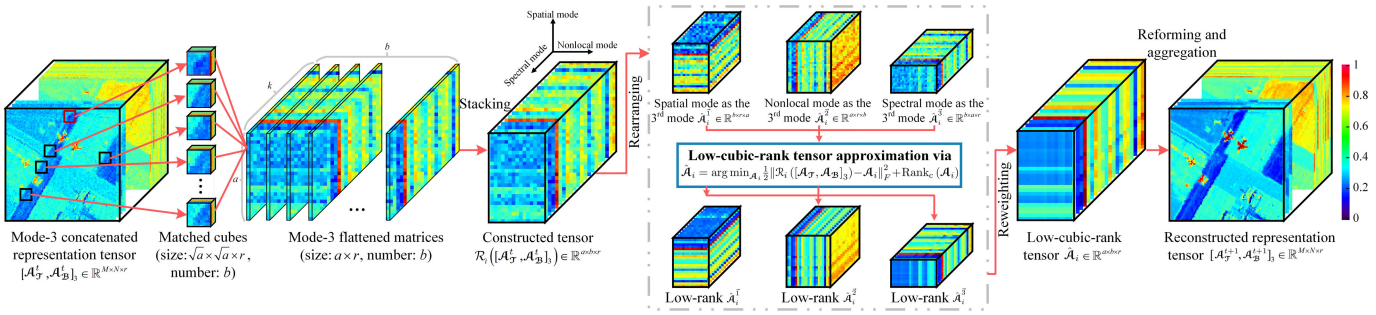


Fig. 4. Illustration of the low-cubic-rank tensor approximation for representation tensors.

with both spatial and spectral degradations, we adopt the alternating direction method of multipliers (ADMM) [81] to optimize each variable. To simplify the optimization procedure, we set auxiliaries $\mathcal{A}=[\mathcal{A}_{\mathcal{T}},\mathcal{A}_{\mathcal{B}}]_3\in\mathbb{R}^{M\times N\times r}$ and $\mathbf{E}=[\mathbf{E}_{\mathcal{T}},\mathbf{E}_{\mathcal{B}}]\in\mathbb{R}^{L\times r}$, transform matrices $\mathbf{D_I}=[\mathbf{O},\mathbf{I}_{r_2}]\in\mathbb{R}^{r_2\times r}$ and $\tilde{\mathbf{D}}_{\mathbf{I}}=[\mathbf{I}_{r_1},\mathbf{O}]\in\mathbb{R}^{r_1\times r}$, and separate $\mathcal{D}_{\text{spa}}$ into the spatial blurring operator $\mathcal{D}_{\mathcal{G}}$ and spatial downsampling operator $\mathcal{D}_{\mathcal{H}}$. Then, we decouple the data fidelity terms and the regularization terms by further introducing auxiliaries $\mathcal{P}_1=\mathcal{D}_{\mathcal{G}}(\mathcal{A})$, $\mathcal{P}_2=\mathcal{P}_3=\mathcal{P}_4=\mathcal{A}$, and $\mathcal{Q}_j=\mathcal{A}_i^{\vec{j}}(j\in[1,3])$. Accordingly, the solution to the model presented in (21) is equal to minimizing the following augment Lagrangian function:

$$\begin{aligned}&\Gamma\left(\mathcal{A},\mathcal{A}_i,\mathcal{P}_k,\mathcal{Q}_j,\mathcal{M}_k,\mathcal{N}_j\right)\\&=\frac{1}{2}\|\mathcal{X}-\mathcal{D}_{\mathcal{H}}(\mathcal{P}_1\times_3\mathbf{E})\|_F^2+\frac{1}{2}\left\|\mathcal{Y}-\mathcal{D}_{\text{spe}}(\mathcal{P}_2\times_3\mathbf{E})\right\|_F^2\\&\quad+\lambda\sum_i\left(\frac{\beta_1}{2}\|\mathcal{R}_i(\mathcal{P}_3)-\mathcal{A}_i\|_F^2+\sum_{j=1}^3\omega_j\text{Rank}_{\text{a}}\left(\mathcal{Q}_j\right)\right.\\&\qquad\left.+\sum_{j=1}^3\frac{\alpha_j}{2}\left\|\mathcal{A}_i^{\vec{j}}-\mathcal{Q}_j-\mathcal{N}_j\right\|_F^2\right)\\&\quad+\gamma\,\Phi_{\text{CEMloss}}\left(\mathcal{P}_4\times_3\mathbf{D_I}\times_3\mathbf{D}_{\mathbf{I}}^T\times_3\mathbf{E}\right)\\&\quad+\frac{\beta_2}{2}\left\|\mathcal{D}_{\mathcal{G}}(\mathcal{A})-\mathcal{P}_1-\mathcal{M}_1\right\|_F^2+\sum_{k=2}^4\frac{\beta_2}{2}\|\mathcal{A}-\mathcal{P}_k-\mathcal{M}_k\|_F^2\end{aligned}\tag{22}$$

where $\mathcal{M}_k$, $\mathcal{N}_j$ $(k\in[1,4],\ j\in[1,3])$ are Lagrange multipliers, and α_j,β_1,β_2 are positive penalty scalars.

Next, we alternatively minimize variables $\mathcal{Q}_j$, $\mathcal{A}_i$, $\mathcal{A}$, $\mathcal{P}_1$, $\mathcal{P}_2$, $\mathcal{P}_3$, $\mathcal{P}_4$ with the others fixed.

1) Updating $\mathcal{Q}_j$: The $\mathcal{Q}_j$ subproblem is composed of three low average rank tensor approximation problems, whose closed-form solutions can be given by a tensor singular value hard thresholding under $\mathcal{L}$ ($\text{THT}_{\mathcal{L}}$) function [78]

$$\mathcal{Q}_j=\mathcal{D}_{\text{THT}_{\mathcal{L}}}\left(\mathcal{A}_i^{\vec{j}}-\mathcal{N}_j,\tau_j\right)\tag{23}$$

where $\mathcal{D}_{\text{THT}_{\mathcal{L}}}(\bullet,\tau)$ is the $\text{THT}_{\mathcal{L}}$ operator with threshold τ, and $\tau_j=\omega_j/\alpha_j$ is the scaled parameter. Note that any real orthogonal transform can be used as $\mathcal{L}$. In this work, we chose discrete cosine transform (DCT) which is more suitable for HSI processing as indicated in [82]. The details of computing $\mathcal{D}_{\text{THT}_{\mathcal{L}}}(\mathcal{X},\tau)$ are presented in Appendix B of the Supplementary.

2) Updating $\mathcal{A}_i$: The $\mathcal{A}_i$ subproblem is a quadratic optimization problem with the solution as follows:

$$\mathcal{A}_i=\left(\beta_1\mathcal{R}_i(\mathcal{P}_3)+\sum_{j=1}^3\alpha_j\left(\mathcal{Q}_j+\mathcal{N}_j\right)\right)./\left(\beta_1+\sum_{j=1}^3\alpha_j\right).\tag{24}$$

3) Updating $\mathcal{A}$: The Plancherel's theorem shows that the sum of the square of a function equals the sum of the square of its Fourier transform. The computation in the frequency domain will be more efficient; thus, we use the 3-D fast Fourier transform $\mathcal{F}$ to derive the following analytical solution for the $\mathcal{A}$ subproblem:

$$\mathcal{A}=\mathcal{F}^{-1}\left(\frac{\mathcal{F}^*\left(\mathcal{D}_{\mathcal{G}}\right)\circ\mathcal{F}(\mathcal{P}_1+\mathcal{M}_1)+\sum_{k=2}^4\mathcal{F}(\mathcal{P}_k+\mathcal{M}_k)}{\mathcal{F}\left(\mathcal{D}_{\mathcal{G}}\right)\circ\mathcal{F}^*\left(\mathcal{D}_{\mathcal{G}}\right)+3\mathcal{I}}\right)\tag{25}$$

where $\circ$ denotes the elementwise product, $\mathcal{F}^{-1}$ and $\mathcal{F}^*$ are the inverse and complex conjugate of $\mathcal{F}$.

4) Updating $\mathcal{P}_1$: The $\mathcal{P}_1$ subproblem can also be solved efficiently in the frequency domain

$$\mathcal{P}_1=\mathcal{D}_{\mathcal{H}}\left(\left(\mathcal{X}\times_3\mathbf{E}^T+\beta_2\mathcal{P}_1'\right)\times_3\mathbf{E}'\right)+\bar{\mathcal{D}}_{\mathcal{H}}\left(\mathcal{P}_1'\right)\tag{26}$$

where $\mathcal{P}_1'=\mathcal{F}^{-1}((\mathcal{F}(\mathcal{D}_{\mathcal{G}})\circ\mathcal{F}(\mathcal{A})-\mathcal{F}(\mathcal{M}_1)))$, $\mathbf{E}'=(\mathbf{E}^T\mathbf{E}+\beta_2\mathbf{I})^{-1}$, and $\bar{\mathcal{D}}_{\mathcal{H}}$ is the complement of $\mathcal{D}_{\mathcal{H}}$.

5) Updating $\mathcal{P}_2$: The $\mathcal{P}_2$ subproblem is a quadratic optimization problem with its closed-form solution given by

$$\begin{aligned}\mathcal{P}_2=&\left(\mathcal{D}_{\text{spe}}^T(\mathcal{Y})\times_3\mathbf{E}^T+\beta_2(\mathcal{A}-\mathcal{M}_2)\right)\\&\times_3\left(\left(\mathcal{D}_{\text{spe}}\mathbf{E}\right)^T\left(\mathcal{D}_{\text{spe}}\mathbf{E}\right)+\beta_2\mathbf{I}\right)^{-1}\end{aligned}\tag{27}$$

where $\mathcal{D}_{\text{spe}}^T$ denotes the transposed spectral response function (SRF).

6) Updating $\mathcal{P}_3$: The $\mathcal{P}_3$ subproblem has the following closed-form solution:

$$\mathcal{P}_3=\left(\lambda\sum_i\mathcal{R}_i^T(\mathcal{A}_i)+\beta_2(\mathcal{A}-\mathcal{M}_3)\right)./\left(\lambda\sum_i\mathcal{R}_i^T\mathcal{R}_i+\beta_2\mathcal{I}\right)\tag{28}$$

where $\mathcal{R}_i^T\mathcal{R}_i$ is the number of overlapping cubes that cover the pixel location i.

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Algorithm 2 Solver for Eq. (14) in Line 5 of Algorithm 1

Require: LR-HSI $\mathcal{X}$, HR-HSI $\mathcal{Y}$, prior spectrum $\mathbf{d}$, spectral bases $\mathbf{E}_{\mathcal{T}}$ and $\mathbf{E}_{\mathcal{B}}$.

Initialization: $\mathcal{A}^0$, $\mathcal{P}_k^0$, $\mathcal{Q}_j^0$, $\mathcal{M}_k^0$, $\mathcal{N}_j^0$, ($k \in [1,4]$, $j \in [1,3]$); Set a, b, λ, γ, $\tau = [\tau_1, \tau_2, \tau_3]$, $\alpha = [\alpha_1, \alpha_2, \alpha_3]$, $\beta_1 = 10^{-4}$, $\beta_2 = 0.05$, $t = 0$, T, and ε.
while not converged **do**
  if $\mathrm{mod}(t, C) = 0$ **then**
    Update the construction of self-correlated tensors;
    Low-cubic-rank tensors approximation: Update $\mathcal{Q}_j^{t+1}$ and $\mathcal{A}_i^{t+1}$ via Eq. (23) and Eq. (24), respectively.
  end if
  Representation tensor learning: Update $\mathcal{A}^{t+1}$ via Eq. (25).
  Spatial super-resolution: Update $\mathcal{P}_1^{t+1}$ via Eq. (26).
  Spectral super-resolution: Update $\mathcal{P}_2^{t+1}$ via Eq. (27).
  Image reconstruction: Update $\mathcal{P}_3^{t+1}$ via Eq. (28).
  Background suppression: Update $\mathcal{P}_4^{t+1}$ via Eq. (29).
  Update Lagrange multipliers $\mathcal{M}_k^{t+1}$, $\mathcal{N}_j^{t+1}$ via Eq. (30).
  Check the convergence: $t > T$ or functional energy $< \varepsilon$.
  Update iterations: $t = t + 1$.
end while
Compute $\mathcal{A}_{\mathcal{T}}$ and $\mathcal{A}_{\mathcal{B}}$ via Eq. (31).

Ensure: Representation tensors $\mathcal{A}_{\mathcal{T}}$ and $\mathcal{A}_{\mathcal{B}}$.

7) Updating $\mathcal{P}_4$: Taking the derivative with respect to $\mathcal{P}_4$ and set the derivative to 0, then the $\mathcal{P}_4$ subproblem can be solved by

$$\mathcal{P}_4 = \mathcal{A} - \mathcal{M}_4 - \frac{\gamma}{\beta_2}\mathtt{fold}_3\left(\frac{\mathbf{D_I}\mathbf{D_I}^T\mathbf{E}^T(\mathbf{R}+\beta\mathbf{I})^{-1}\mathrm{Re}(\mathbf{d})}{\mathbf{d}^T(\mathbf{R}+\beta\mathbf{I})^{-1}\mathbf{d}}\right) \tag{29}$$

where $\mathtt{fold}_3(\bullet)$ denotes the matrix mode-3 folding operation.

8) Updating $\mathcal{M}_k$, $\mathcal{N}_j$: Finally, the update of the Lagrange multipliers is given by

$$\begin{cases} \mathcal{M}_k = \mathcal{M}_k - (\mathcal{M}' - \mathcal{P}_k) \\ \mathcal{N}_j = \mathcal{N}_j - (\mathcal{A}_i - \mathcal{Q}_j) \end{cases} \tag{30}$$

where $\mathcal{M}' = \mathcal{D}_{\mathcal{G}}(\mathcal{A})$ for $k = 1$, and $\mathcal{M}' = \mathcal{A}$ for $k \in [2,4]$.

When the iteration terminates, the representation tensors $\mathcal{A}_{\mathcal{T}}$ and $\mathcal{A}_{\mathcal{B}}$ are computed by

$$\begin{cases} \mathcal{A}_{\mathcal{T}} = \mathcal{A} \times_3 \tilde{\mathbf{D}}_{\mathbf{I}} \\ \mathcal{A}_{\mathcal{B}} = \mathcal{A} \times_3 \mathbf{D}_{\mathbf{I}}. \end{cases} \tag{31}$$

The complete procedure of solving (14) by optimizing (21) is summarized in Algorithm 2, where the functional energy is $\|\mathcal{A}^{t+1} \times_3 \mathbf{E} - \mathcal{A}^t \times_3 \mathbf{E}\|_F^2 / \|\mathcal{A}^t \times_3 \mathbf{E}\|_F^2$, and an integer C is used to control the number of cube matching [83].

IV. Numerical Experiments

In this section, we report the experimental results for both tasks. Due to space limitations in this article, we have provided more results in Appendix C of the Supplementary.

A. Experimental Configurations

1) Datasets: Five commonly used public datasets containing target truth were employed to quantitatively verify the effectiveness of our HySRTD method, i.e., Sandiego I,[1] Sandiego II,[1] Gulfport,[2] Los Angeles,[2] and Segundo [66]. The Sandiego I and II datasets have 186 available spectral bands after removing the bands damaged by water absorption. Gulfport and Los Angeles datasets have 191 and 205 high-quality spectral bands, respectively. All of them contain a region of 100 × 100 spatial scenes and several airplanes are considered targets. The Segundo dataset has 220 spectral bands and contains a larger region of 250 × 250 spatial scenes in which the constructions of oil refineries, such as towers and storage tanks, are considered targets. Finally, the real Houston2018[3] dataset, which does not contain the reference HR-HSI and target truth, was used to evaluate the performance in the typical real-world HSI super-resolution and target detection scenario.

2) Baselines: For the HSI super-resolution task, the competitors included state-of-the-art model optimization-based methods (i.e., sparse representation method NSSR [14], matrix factorization method RLRMDF [22], tensor representation methods NPTSR [19], LTMR [34], and TenSR [35]) and learning-based methods (i.e., convolutional neural network CNNFUS [37], coupled autoencoder network HyCoNet [43], and mutual Dirichlet network u2MDN [45]).

For the HSI target detection task, we also compared the latest developments: collaborative sparse representation-based method CSCR [60], hierarchical suppression-based method hCEM [51], rank minimization-based method LRLS [59], sparse and low-rank decomposition method SLRMD [58], convolutional neural network HTD-Net [67], and background learning-based autoencoder network BLTSC [66].

All codes were downloaded from the authors' homepages, and the parameters were fine-tuned as suggested in the reference papers to achieve optimal results.

3) Implementation Details: For the HSI super-resolution task, the four-band HR-MSI was generated by averaging the wavelength range of the reference HR-HSI according to the SRF of the R.G.B. and NIR bands of the IKONOS multispectral sensor as is widely used in remote sensing. Meanwhile, the spatial scaling factor (sf) for generating the downsampled LR-HSI was set as sf = 5 and sf = 10. Before these two cases of spatial scaling, the Gaussian blurring of size 9 × 9 and 19 × 19 with standard deviations 2.12 and 4.24 was applied to all bands of the reference HR-HSI, respectively. For the HSI target detection task, only a single prior spectrum was required, which is obtained by averaging five spectra randomly selected from the target pixels. PSNR, structure similarity (SSIM), cross correlation (CC), and SAM were employed to quantitatively evaluate the super-resolution results. The receiver operating characteristic (ROC) curves [84] and the area under the ROC curve (AUC) were employed to evaluate the detection results. The closer the ROC curve lies in the

[1] https://aviris.jpl.nasa.gov/data/index.html
[2] http://xudongkang.weebly.com/
[3] https://hyperspectral.ee.uh.edu/?page_id=1075

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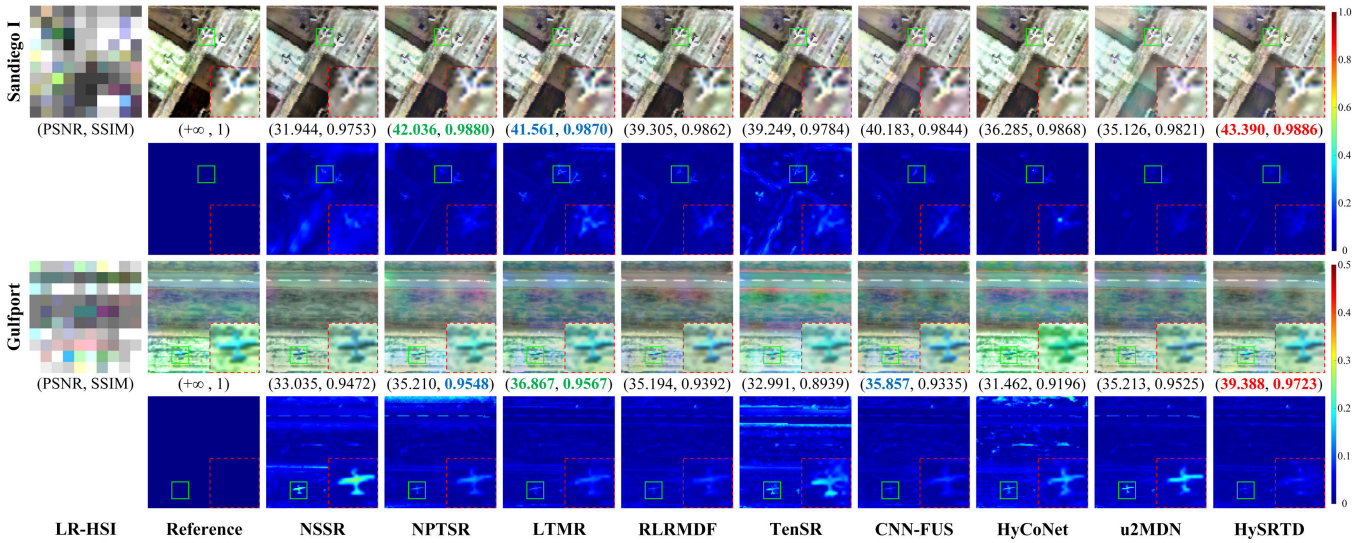


Fig. 5. Comparison of pseudo-color images (R: 37, G: 17, B: 7) and residual error maps of super-resolution results with an sf of 10. (Hereinafter, the first to third best quantitative results are highlighted in **RED**, **GREEN**, and **BLUE**, respectively.)

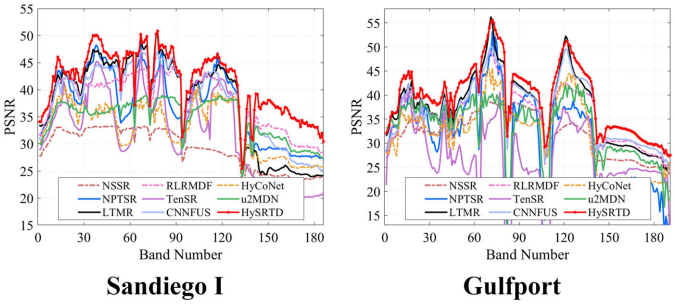


Fig. 6. PSNR as a function of spectral bands with an sf of 10.

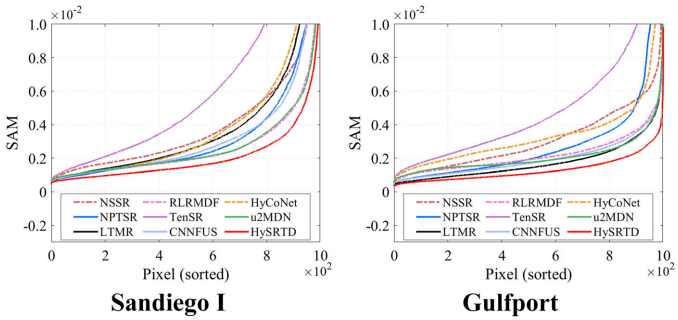


Fig. 7. SAM (plotted on a $\log_{10}(\bullet)$ scale) as a function of sorted pixels with an sf of 10.

upper left corner and the larger the AUC value, the better the detection rate. Before evaluations, the gray values of each band were normalized to the interval [0, 1]. Due to the presence of two layers of iterative solution in our method, it is unnecessary to laboriously obtain the exact solutions in Algorithm 2. That is, we only perform the construction of self-correlated tensors based on Euclidean distance [85], [86] once each time we enter line 5 of Algorithm 1, thereby achieving a trade-off between efficiency and accuracy.

B. Results on HSI Super-Resolution Task

Fig. 5 intuitively presents the visual comparison of super-resolution results on the heavily degraded case of sf = 10 of the Sandiego I and Gulfport datasets. More results on the Sandiego II, Los Angeles, and Segundo datasets are presented in Appendix C-A. Mean PSNR and SSIM results of pseudo-color bands, residual error maps, and meaningful regions marked and enlarged three times are further presented for better visibility. We can observe that the proposed HySRTD obtained the clearest pseudo-color map, the highest quantitative results, and the smoothest residual map. Table I reports quantitative results obtained by all methods on all five datasets, where our HySRTD method consistently outperforms other competing ones in almost all metrics. Deep learning-based CNNFUS and u2MDN, subspace decomposition-based LTMR, and matrix factorization-based RLRMDF have all achieved competitive results. Similar to the proposed HySRTD, these methods exploited the low-dimensional spectral structure of HSI. We further evaluate the pros and cons of different methods using band-by-band PSNR and pixel-by-pixel SAM curves. As shown in Fig. 6, our HySRTD demonstrated a significant advantage in most bands, especially for the end of near-infrared bands that are not covered in SRF. This phenomenon verified that the proposed low-cubic-rank regularization fully characterizes the implicit correlations of the HSI tensor along different modes. Meanwhile, as shown in Fig. 7, HySRTD achieved the average best in terms of spectral fidelity across all pixels, which highlights the effectiveness of the refinement procedure of spectral bases.

C. Results on HSI Target Detection Task

Here, we test our HySRTD method on the detection task. Note that in our method, target detection results are output with super-resolution results concurrently. The visual detection maps are shown in Fig. 8. It can be observed that even under such an unfair comparison, the proposed HySRTD still delivered the most precise target

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TABLE I

COMPARISON OF QUANTITATIVE RESULTS OBTAINED BY DIFFERENT SUPER-RESOLUTION METHODS ON ALL DATASETS

Dataset	sf	Index	NSSR [14]	NPTSR [19]	LTMR [34]	RLRMDF [22]	TenSR [35]	CNNFUS [37]	HyCoNet [43]	u2MDN [45]	HySRTD
Sandiego I	5	PSNR↑	30.979	38.491	**39.796**	39.705	34.653	**40.072**	33.256	36.341	**43.223**
		SSIM↑	0.9472	**0.9752**	0.9630	0.9738	0.9037	**0.9746**	0.9592	0.9718	**0.9829**
		CC↑	0.9923	0.9976	0.9971	**0.9984**	0.9914	**0.9985**	0.9955	0.9969	**0.9991**
		SAM↓	0.0285	0.0250	0.0272	**0.0226**	0.0463	**0.0215**	0.0285	0.0272	**0.0178**
	10	PSNR↑	29.065	36.813	**37.581**	**37.721**	31.648	37.570	32.877	34.558	**41.411**
		SSIM↑	0.9448	**0.9708**	0.9368	0.9675	0.8528	0.9598	0.9437	**0.9703**	**0.9788**
		CC↑	0.9874	0.9961	0.9935	**0.9974**	0.9808	0.9959	0.9937	**0.9970**	**0.9987**
		SAM↓	0.0387	0.0328	0.0401	**0.0262**	0.0704	0.0334	0.0402	**0.0270**	**0.0207**
Sandiego II	5	PSNR↑	33.088	38.825	**41.753**	40.081	36.940	**40.366**	33.743	37.578	**43.709**
		SSIM↑	0.9534	**0.9714**	0.9683	0.9682	0.9233	0.9695	0.9581	**0.9708**	**0.9790**
		CC↑	0.9912	0.9955	**0.9972**	**0.9974**	0.9917	0.9971	0.9928	0.9965	**0.9985**
		SAM↓	0.0243	0.0220	**0.0196**	**0.0196**	0.0309	**0.0193**	0.0240	0.0222	**0.0156**
	10	PSNR↑	31.109	37.279	**39.076**	**38.107**	34.313	37.993	33.351	36.388	**42.510**
		SSIM↑	0.9464	0.9613	0.9317	**0.9621**	0.8826	0.9534	0.9413	**0.9661**	**0.9732**
		CC↑	0.9855	0.9930	0.9908	**0.9961**	0.9815	0.9916	0.9864	**0.9946**	**0.9975**
		SAM↓	0.0325	0.0273	0.0335	**0.0240**	0.0451	0.0310	0.0406	**0.0252**	**0.0195**
Gulfport	5	PSNR↑	33.749	32.851	**38.322**	38.317	29.813	**38.938**	34.036	36.691	**40.384**
		SSIM↑	0.9809	0.9746	**0.9869**	0.9840	0.9336	**0.9844**	0.9784	**0.9884**	**0.9889**
		CC↑	0.9884	0.9482	0.9926	**0.9930**	0.9500	**0.9932**	0.9888	0.9880	**0.9953**
		SAM↓	0.0217	0.0197	**0.0151**	0.0163	0.0323	**0.0152**	0.0214	0.0153	**0.0121**
	10	PSNR↑	31.869	31.057	**37.365**	35.821	26.944	**36.897**	32.009	33.172	**39.649**
		SSIM↑	0.9746	0.9646	**0.9823**	0.9791	0.8944	0.9787	0.9576	**0.9792**	**0.9884**
		CC↑	0.9835	0.9462	**0.9905**	0.9887	0.9317	**0.9895**	0.9786	0.9687	**0.9942**
		SAM↓	0.0307	0.0290	**0.0176**	0.0221	0.0500	**0.0199**	0.0346	0.0208	**0.0131**
Los Angeles	5	PSNR↑	32.787	36.698	**39.206**	**38.418**	32.540	37.224	34.784	35.289	**40.295**
		SSIM↑	0.9767	**0.9862**	**0.9911**	0.9860	0.9589	0.9838	0.9853	0.9853	**0.9911**
		CC↑	0.9639	0.9697	**0.9784**	**0.9745**	0.9458	0.9699	0.9738	0.9671	**0.9831**
		SAM↓	0.0322	**0.0231**	**0.0179**	0.0243	0.0350	0.0254	0.0273	0.0247	**0.0175**
	10	PSNR↑	31.481	32.576	**37.701**	**37.146**	29.382	36.066	32.901	34.392	**38.699**
		SSIM↑	0.9766	0.9686	**0.9874**	**0.9841**	0.9261	0.9795	0.9792	0.9823	**0.9886**
		CC↑	0.9624	0.9573	**0.9723**	**0.9690**	0.9143	0.9619	0.9667	0.9538	**0.9763**
		SAM↓	0.0475	0.0333	**0.0228**	0.0294	0.0528	0.0308	0.0394	**0.0284**	**0.0204**
Segundo	5	PSNR↑	33.527	34.165	**37.031**	**36.290**	31.107	35.048	33.806	35.812	**38.424**
		SSIM↑	0.9628	0.9554	**0.9782**	0.9621	0.8963	0.9535	0.9686	**0.9694**	**0.9740**
		CC↑	0.9124	0.8911	**0.9311**	**0.9241**	0.8612	0.9171	0.8989	0.9108	**0.9434**
		SAM↓	0.0496	0.0422	**0.0326**	0.0448	0.0663	0.0577	0.0401	**0.0350**	**0.0342**
	10	PSNR↑	32.548	31.584	**35.074**	**34.750**	29.161	34.425	32.986	34.378	**37.700**
		SSIM↑	0.9531	0.9179	**0.9672**	**0.9607**	0.8429	0.9534	0.9574	0.9556	**0.9693**
		CC↑	0.8779	0.8700	**0.9038**	0.9029	0.8352	**0.9140**	0.8809	0.8897	**0.9326**
		SAM↓	0.0585	0.0560	**0.0418**	0.0518	0.0885	0.0673	0.0549	**0.0427**	**0.0367**

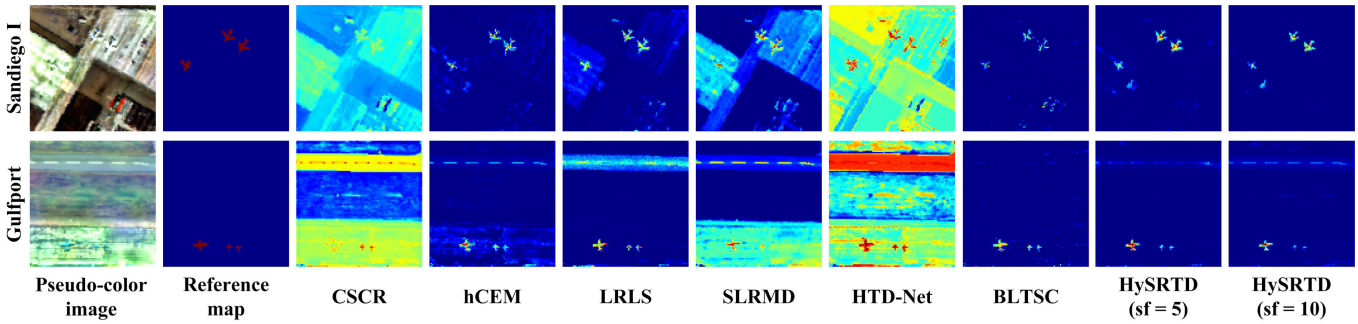


Fig. 8. Two-dimensional plots of the visual detection maps obtained by different target detection methods. Note that the input for all comparison methods is the untouched reference image and the inputs for our method are simply the downsampled data pairs with the sfs of 5 and 10.

distribution with superior background suppression. The background learning-based BLTSC obtained suboptimal results because it also pays attention to the background part. The quantitative ROC curves obtained by all methods on the reference HSI and our method on two scaling cases are drawn in Fig. 9. The proposed HySTRD achieved the optimal ROC curve on Los Angeles and Segundo datasets under both sf $= 5$ and sf $= 10$ cases. That is, the ROC curve obtained by HySTRD perfectly wrapped the curves obtained by the competitors. For the other datasets, HySTRD also converged fastest to a probability of detection 1 and demonstrated a very competitive detection ability, which demonstrates that the proposed method is indeed effective for the HSI target detection task. More visual maps and ROC results are presented in Appendix C-B. For some datasets that are difficult to distinguish the ROC curves clearly, we further report the quantitative AUC results in Table II. Compared with all competitors that require the reference HR-HSI, the proposed HySRTD achieves the average optimal AUC values when sf $= 5$. When sf $= 10$, it also achieves the top three best AUC values. Some other competitors can detect the targets well, but it is difficult for them to suppress the background well. For instance, the hierarchical-based method hCEM achieved a slightly higher AUC than HySRTD on Gulfport. However, as shown in the third row of Fig. 8, the background suppression of hCEM is inferior to our HySRTD.

D. Real HSI Super-Resolution and Target Detection

We then verify the proposed HySRTD method in the real-world HSI super-resolution and target detection scenario.

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TABLE II
COMPARISON OF AUC RESULTS OBTAINED BY DIFFERENT TARGET DETECTION METHODS ON ALL DATASETS

Dataset	CSCR	hCEM	LRLS	SLRMD	HTD-Net	BLTSC	HySRTD (sf = 5)	HySRTD (sf = 10)
Sandiego I	0.9826	0.9797	**0.9962**	0.9503	0.8981	0.9192	**0.9980**	**0.9967**
Sandiego II	**0.9965**	0.9855	0.9845	0.9917	0.8833	0.9891	**0.9977**	**0.9961**
Gulfport	0.8662	**0.9977**	0.9802	0.9594	0.9860	0.9704	**0.9974**	**0.9927**
Los Angeles	0.8394	**0.9222**	0.7888	0.8404	0.7174	0.8771	**0.9931**	**0.9922**
Segundo	0.8457	0.8774	0.9092	**0.9859**	0.9739	0.8395	**0.9942**	**0.9804**

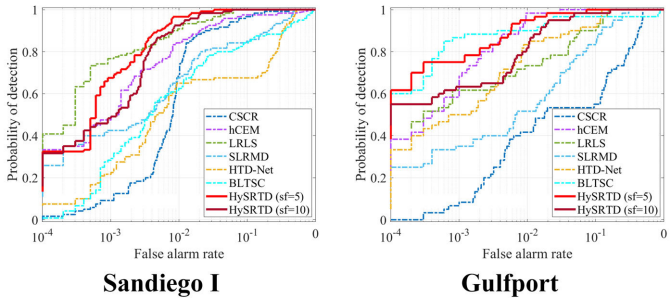


Fig. 9. ROC curves obtained by different target detection methods.

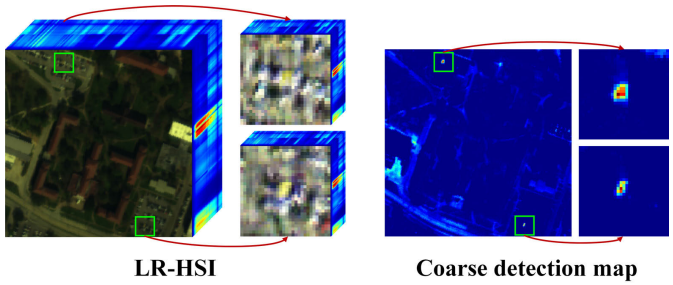


Fig. 10. Two regions in the real Houston2018 dataset with high probability of containing vehicle targets of interest.

The employed Houston2018 dataset that provides LR-HSI ($596 \times 601 \times 48$) and HR-MSI ($11920 \times 12\,020 \times 3$) data pairs have an sf of 20, while the exact blurring kernel and SRF are unacquirable. Such harsh conditions crippled many super-resolution methods, further limiting subsequent high-level vision tasks. We first performed a coarse detection on LR-HSI using the CEM detector. As shown in Fig. 10, two vehicles were considered targets, and there were two regions with a very high probability of targets present. However, due to the limited spatial resolution, it is extremely difficult to further discriminate the target information from the sampled LR-HSI and detection map. When confronted with application scenarios characterized by elevated conditional risk, such detection outcomes are difficult to accept. Therefore, we compared the super-resolution and target detection results in these two regions. The downsample procedures were estimated using the method in [24]. Considering the practicability in real applications, only one pixel with the highest energy in the right column of Fig. 10 was selected as the prior spectrum for HySRTD. As shown in Fig. 11, under such an extreme scaling case, most competing methods failed to deliver the correct spatial information or caused a terrible spectral distortion, even ones that performed strikingly in synthetic experiments, e.g., LTMR and CNNFUS. Our HySRTD delivered the best super-resolution results and accurate detection maps with more rich target information. The success of the real experiment verified the practical utility of our novel unified solution.

TABLE III
EFFECTIVENESS OF THE PROPOSED TENSOR LOW-CUBIC-RANK REGULARIZATION

Rearranged third mode			Sandiego I		Gulfport	
spatial	nonlocal	spectral	PSNR	SAM	PSNR	SAM
✓			40.190	0.0239	37.187	0.0179
	✓		40.732	0.0217	37.882	0.0164
		✓	39.443	0.0231	37.607	0.0165
✓	✓		40.680	0.0216	37.798	0.0165
✓		✓	**40.818**	**0.0215**	**38.985**	**0.0149**
	✓	✓	**41.379**	**0.0208**	**39.568**	**0.0132**
✓	✓	✓	**41.411**	**0.0207**	**39.649**	**0.0131**

V. ABLATION STUDY

A. Effectiveness of Tensor Low-Cubic-Rank Regularization

The proposed low-cubic-rank regularization ensures the HSI super-resolution results and provides more accurate information for target detection by considering all the rearranging patterns of the constructed tensors. Here, we investigate the effectiveness of this new tensor rank measure on Sandiego I and Gulfport datasets. Table III reports the quantitative PSNR and SAM results obtained by considering different modes as the third mode of the rearranged constructed tensors. We can observe that considering all three modes always delivers the best fusion results both spatially and spectrally. Meanwhile, considering both the non-local and spectral modes leads to the second-best. If only a single mode is considered, the best choice is the non-local mode. Therefore, in real applications, if the requirement for parameter number is more strict than the requirement for super-resolution accuracy, we recommend considering the non-local mode, i.e., setting $\alpha = [0, 1, 0]$. If accuracy is the first priority, we recommend considering all three modes. According to the above results, we empirically fixed weights $\alpha = [0.2, 0.7, 0.1]$ in all experiments. As for τ, we recommend selecting in the interval $[5e-3, 5e-1]$ according to the degradation level of different datasets.

B. Impact of Parameters

The practicality of implementation in real-world applications is determined not only by accuracy but also by the sensitivity and complexity of parameters. Here, we analyze the parameters that play a key role in the proposed HySRTD method, i.e., the number of spectral bases r, length of similar cubes $(a)^{1/2}$, number of cubes b, and regularization parameters

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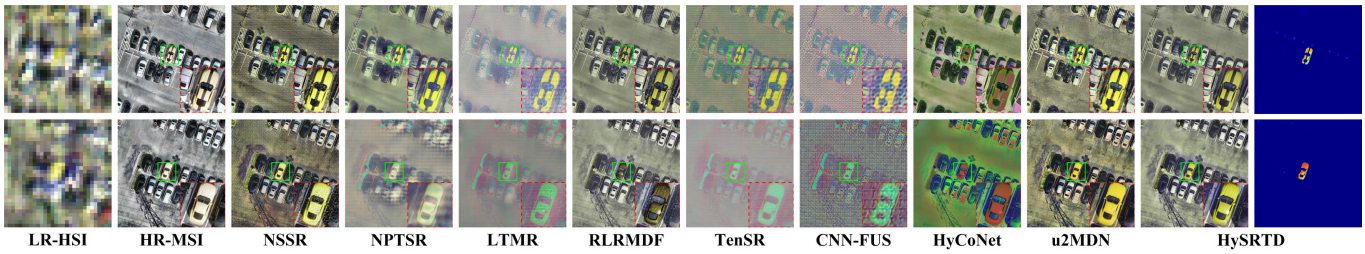


Fig. 11. Comparison of pseudo-color images (R: 20, G: 15, B: 5) of the super-resolution results with an sf of 20 for two regions of the real Houston2018 dataset. The proposed HySRTD method not only offers superior spatial enhancement and spectral fidelity, but also provides significantly more accurate target maps compared with the coarse detection results.

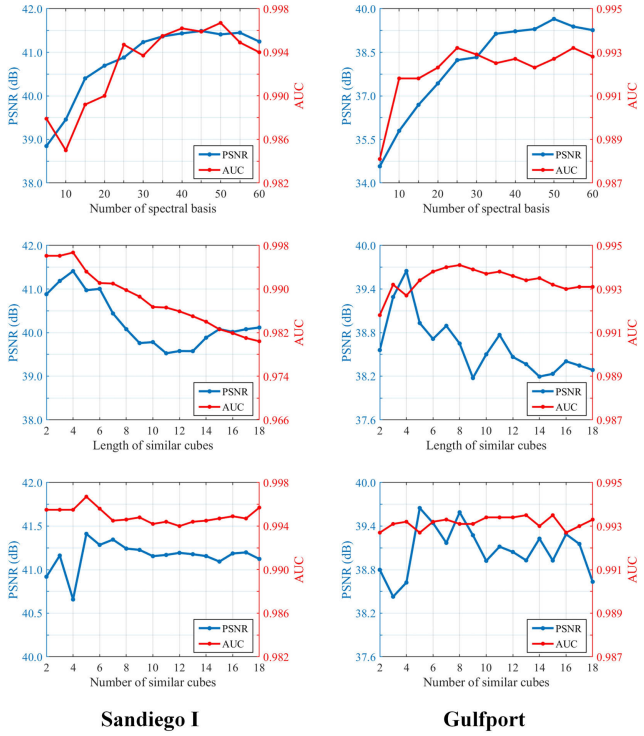


Fig. 12. Sensitivity analysis of the model parameters.

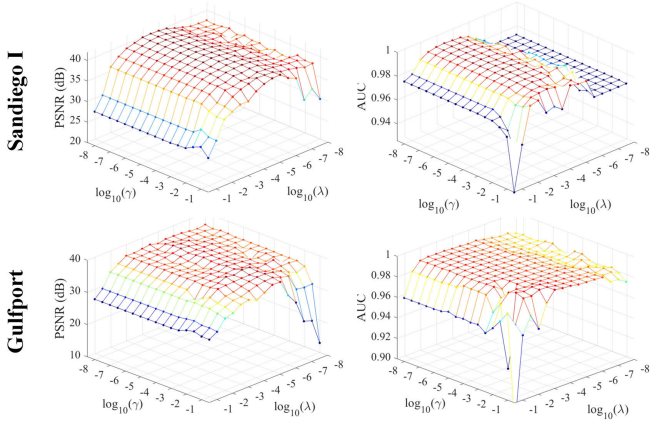


Fig. 13. Sensitivity analysis of the regularization parameters.

TABLE IV

COMPARISON OF EXECUTION EFFICIENCY OF ALL SUPER-RESOLUTION METHODS ON SANDIEGO I DATASET (IN SECONDS)

NSSR	NPTSR	LTMR	RLRMDF	TenSR	CNNFUS	HyCoNet	u2MDN	HySRTD
44.43	68.83	28.72	10.18	257.21	6.52	696.02	392.39	41.73

λ, γ. Since the parameters collectively influence the performance of both super-resolution and target detection tasks, we employ metrics PSNR and AUC as evaluative measures. From the first column of Fig. 12, we can observe that as r increases, the accuracy of both tasks gradually increases and tend to be stable. Therefore, we fixed $r = 50$ in all experiments. Likewise, the second column of Fig. 12 shows that a small value of $(a)^{1/2}$ is sufficient to deliver satisfying accuracy. Meanwhile, since the $(a)^{1/2}$ determines the size of the spatial mode of constructed tensors, it also has a conspicuous impact on execution efficiency. Therefore, considering the trade-off between accuracy and efficiency, we fixed $(a)^{1/2} = 4$ in all experiments. For b, which has less impact on execution efficiency, we can observe that its impact on accuracy is also limited. Therefore, we fixed $b = 5$ in all experiments. Regularization parameters λ and γ control the proportions of the low-cubic-rank term and CEM loss term, respectively. From Fig. 13, we can see that the sensitivity of λ and γ within a certain range is stable. In other words, fine results can be obtained without a large range of manual parameter searching, which ensures practicality in real-world applications. Therefore, in our experiments, we fixed λ to $1E-4$ in all experiments and set γ to $1E-4$ for Sandiego I and $5E-3$ for Gulfport.

C. Computational Efficiency

The average seconds consumed by all super-resolution methods on Sandiego I are reported in Table IV. All methods were run on a Windows 10 workstation with dual Intel Xeon E5-2640V4@2.4 GHz CPU, an NVIDIA 2080Ti GPU, and 128 GB of RAM. Implementation platforms include MATLAB R2021a, pytorch v1.7, and Tensorflow v2.0. Parallel computing can be used to realize a fast process of the constructed tensors [85]. It can be seen that even though both the super-resolution and target detection tasks are realized simultaneously, the proposed HySRTD still achieved competitive operational efficiency compared with the state-of-the-art ones.

D. Convergence Analysis

The convergence of the multiblock ADMM has been extensively discussed [81], [87]. In Algorithm 2, the convergence

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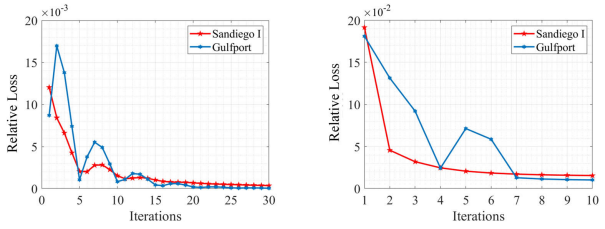


Fig. 14. Empirical convergence behaviors of the (right) Algorithm 1 and (left) Algorithm 2.

conditions of the low-rank approximation subproblems have been discussed in study [78], while the remaining subproblems have Lipschitz continuous gradients. Moreover, by decoupling the solutions to spectral bases and representation tensors in Algorithm 1, a sophisticated optimization procedure is unnecessary, and a simple alternating update scheme converges satisfactory results. To support this claim, we provide an empirical convergence analysis. Fig. 14 depicts the decreasing trend of the relative loss of the functional energies. It is evident that the loss curves stabilize as the number of iterations increases, validating the practical convergence of our algorithms.

VI. Conclusions and Future Work

In this work, we commit to opening the door for a practical and pressing problem: how to mutually optimize both the super-resolution and target detection tasks for the degraded HSI data. Consequently, we introduced a unified solution that brings new insights for HSI processing communities. We overcame the obstacle of connecting HSI low-level and high-level visions by constructing different spectral bases for target and background subspaces. To comprehensively leverage the inherent properties of HSI, we introduced a novel tensor cubic rank and showed the possibility of turning the well-established theory into a loss term. Jointly optimizing both tasks helps alleviate the additional errors, redundancies, and inefficiencies associated with previous practices. Extensive evaluations and analyses have substantiated the effectiveness and superiority of the proposed method. The success of this endeavor would largely facilitate the multitask applications of HSI in real-world scenarios.

In future work, we plan to expand the scope of our unified solution to include more low-level vision tasks and to incorporate more high-level vision tasks. Additionally, developing high-order tensorized models with more sophisticated and appropriate HSI priors is always interesting.

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